# MAST30027 Modern Applied Statistics

Mock Paper – Worked Solutions and Mark Scheme

Semester 2, 2026 (9 questions, 78 marks)

**Conventions.** Every answer is rounded to 2 d.p. unless the question says otherwise and shown in a box. "FX-82" lines say what to key in. Intermediate values are carried at full calculator precision and displayed to 4–5 d.p. Gamma$(a, b)$ means shape $a$, rate $b$. The **Marks** lines show where each mark is earned; follow-through marks are given for correct method on an earlier wrong value.

## Question 1 [7 marks]

**(a)** *Log-likelihood of $\theta$ in terms of $N_1, N_2, N_3$.* [2]

$$L(\theta) = \big(\theta^2\big)^{N_1}\big(2\theta(1-\theta)\big)^{N_2}\big((1-\theta)^2\big)^{N_3} \qquad \text{(independent)}$$

$$\ell(\theta) = 2N_1\log\theta + N_2\log 2 + N_2\log\theta + N_2\log(1-\theta) + 2N_3\log(1-\theta) \qquad \text{(take logs)}$$

$$\boxed{\ell(\theta) = (2N_1 + N_2)\log\theta + (N_2 + 2N_3)\log(1-\theta) + N_2\log 2}$$

**Marks:** 1 for the product/likelihood in terms of $N_k$; 1 for the collected log-likelihood (constant $N_2\log 2$ may be dropped).

**Common errors:** forgetting the factor 2 in the $Aa$ probability; writing $\theta^{N_1}$ instead of $(\theta^2)^{N_1}$.

**(b)** *Fisher information $I(\theta)$ for the whole sample.* [3]

$$\ell'(\theta) = \frac{2N_1 + N_2}{\theta} - \frac{N_2 + 2N_3}{1-\theta}$$

$$\ell''(\theta) = -\frac{2N_1 + N_2}{\theta^2} - \frac{N_2 + 2N_3}{(1-\theta)^2}$$

$$I(\theta) = -\,\mathbb{E}[\ell''(\theta)] = \frac{\mathbb{E}(2N_1 + N_2)}{\theta^2} + \frac{\mathbb{E}(N_2 + 2N_3)}{(1-\theta)^2} \qquad (-\,\mathbb{E} \text{ of second derivative})$$

$$\mathbb{E}(2N_1 + N_2) = 2n\theta^2 + 2n\theta(1-\theta) = 2n\theta \qquad \text{(given expectations)}$$

$$\mathbb{E}(N_2 + 2N_3) = 2n\theta(1-\theta) + 2n(1-\theta)^2 = 2n(1-\theta)$$

$$I(\theta) = \frac{2n\theta}{\theta^2} + \frac{2n(1-\theta)}{(1-\theta)^2} = \frac{2n}{\theta} + \frac{2n}{1-\theta}$$

$$= \frac{2n\big[(1-\theta) + \theta\big]}{\theta(1-\theta)}$$

$$\boxed{I(\theta) = \frac{2n}{\theta(1-\theta)}} \qquad \text{(common denominator)}$$

**Marks:** 1 for $\ell''$; 1 for substituting expectations correctly; 1 for simplified result.

**Common errors:** using $+\,\mathbb{E}[\ell'']$ (negative information); not simplifying to a single fraction; using $n$ instead of $2n$ (each individual carries two alleles).

**(c)** *MLE, its asymptotic distribution, and values for $n = 100$, $N = (34, 46, 20)$.* [2]

$$\begin{aligned}
\ell'(\hat\theta) = 0 \;\Rightarrow\; (2N_1 + N_2)(1-\hat\theta) &= (N_2 + 2N_3)\hat\theta \\
2N_1 + N_2 &= \hat\theta\,(2N_1 + 2N_2 + 2N_3) && \text{(collect } \hat\theta \text{ terms)} \\
\hat\theta &= \frac{2N_1 + N_2}{2n} && (\textstyle\sum N_k = n;\ \ell'' < 0) \\
\hat\theta &\overset{a}{\sim} \text{Normal}\Big(\theta,\ \frac{1}{I(\theta)}\Big) \\
&= \text{Normal}\Big(\theta,\ \frac{\theta(1-\theta)}{2n}\Big) \\
\hat\theta &= \frac{2(34) + 46}{200} = \frac{114}{200} = \boxed{0.57} \\
\widehat{\text{Var}}(\hat\theta) &= \frac{0.57 \times 0.43}{2 \times 100} = \frac{0.2451}{200} = 0.0012255 = \boxed{0.0012}
\end{aligned}$$

**FX-82:** $0.57 \times 0.43 \div 200 =$ gives $1.2255 \times 10^{-3}$; to 4 d.p. this is 0.0012.

**Marks:** 1 for $\hat\theta$ formula and value 0.57; 1 for the asymptotic normal with variance $\theta(1-\theta)/(2n)$ and the plug-in 0.0012.

**Common errors:** using $n$ rather than $2n$ in the denominator (variance 0.0025); giving the standard deviation (0.035) instead of the variance.

## Question 2 [8 marks]

**(a)** *Write the pmf of $Y = N/m$ in exponential dispersion form; identify $\theta, \phi, a, b, c$.* [4]

$$\begin{aligned}
f(y) = P(N = my) &= \frac{e^{-m\mu}(m\mu)^{my}}{(my)!} && (Y = y \iff N = my) \\
\log f(y) &= my\log\mu - m\mu + my\log m - \log\big((my)!\big) && \text{(take logs)} \\
&= \frac{y\log\mu - \mu}{1/m} + my\log m - \log\big((my)!\big) && \text{(factor out } m = 1/\phi) \\
&= \frac{y\theta - e^\theta}{\phi} + \frac{y}{\phi}\log\frac{1}{\phi} - \log\Big(\big(\tfrac{y}{\phi}\big)!\Big) && (\theta = \log\mu,\ \phi = 1/m)
\end{aligned}$$

Matching with $\dfrac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)$:

$$\boxed{\theta = \log\mu, \quad \phi = \tfrac{1}{m}, \quad a(\phi) = \phi, \quad b(\theta) = e^\theta, \quad c(y,\phi) = \tfrac{y}{\phi}\log\tfrac{1}{\phi} - \log\big(\big(\tfrac{y}{\phi}\big)!\big)}$$

**Marks:** 1 for the pmf of $Y$ via $N = my$; 1 for $\theta = \log\mu$; 1 for $a(\phi) = \phi$ and $b(\theta) = e^\theta$; 1 for $c(y,\phi)$ (equivalently $my\log m - \log((my)!)$).

**Common errors:** leaving $\mu$ in $b$ instead of $e^\theta$; forgetting that $my$ (not $y$) is the factorial argument; putting $m$ inside $\theta$.

**(b)** *Canonical link;* $\mathbb{E}(Y)$, $\mathrm{Var}(Y)$ *from* $b$*; variance function.* [2]

$$\begin{aligned}
\theta = \log\mu \;\Rightarrow\; g(\mu) &= \log\mu && \text{(canonical link: } \eta=\theta\text{)}\\
\mathbb{E}(Y) = b'(\theta) &= e^{\theta} = \mu\\
\mathrm{Var}(Y) = a(\phi)\, b''(\theta) &= \phi\, e^{\theta} = \frac{\mu}{m}\\
v(\mu) = b''(\theta(\mu)) &= \mu && (\mathrm{Var}(Y) = a(\phi)v(\mu))\\
\boxed{g = \log,\quad \mathbb{E}(Y)=\mu,\quad \mathrm{Var}(Y)=\mu/m,\quad v(\mu)=\mu}
\end{aligned}$$

**Marks:** 1 for the log link and $\mathbb{E}(Y)=\mu$; 1 for $\mathrm{Var}(Y)=a(\phi)b''=\mu/m$ and $v(\mu)=\mu$.

**Common errors:** writing $v(\mu)=\mu/m$ (the $1/m$ belongs to $a(\phi)$).

**(c)** *IWLS weight* $W_{ii}$ *and why longer trapping gets more weight.* [2]

$$\begin{aligned}
g(\mu)=\log\mu \;\Rightarrow\; g'(\mu_i) &= \frac{1}{\mu_i}\\
\mathrm{Var}(z_i) = g'(\mu_i)^2\, v(\mu_i)\, a_i &= \frac{1}{\mu_i^2}\cdot\mu_i\cdot\frac{1}{m_i} = \frac{1}{m_i\mu_i}\\
W_{ii} = \frac{1}{\mathrm{Var}(z_i)} &= \boxed{m_i\mu_i}
\end{aligned}$$

Model answer: a trap left out longer counts more insects, so its rate $N_i/m_i$ has variance $\mu_i/m_i$, which is smaller. More precise observations get a larger weight $m_i\mu_i$.

**Marks:** 1 for $W_{ii}=m_i\mu_i$ with working; 1 for the explanation (smaller variance of the rate ⇒ more weight).

**Common errors:** using $a_i=1$ (giving $W=\mu_i$); $g'(\mu)=\mu$.

## Question 3 [9 marks]

**(a)** *Odds ratios for spray and per cm of rain, with interpretation.* [2]

$$\begin{aligned}
\widehat{OR}_{\text{spray}} &= e^{\hat\beta_2} = e^{-0.64442} = \boxed{0.52}\\
\widehat{OR}_{\text{rain}} &= e^{\hat\beta_1} = e^{0.29876} = \boxed{1.35}
\end{aligned}$$

**FX-82:** SHIFT ln gives $e^x$: $e^x$ (−) 0.64442 = gives 0.52497; $e^x$ 0.29876 = gives 1.34819.

Interpretation: at the same rainfall, the odds of a tree being infected in a sprayed plot are 0.52 times those in an unsprayed plot (about 48% lower). Each extra cm of monthly rain multiplies the odds of infection by 1.35 (a 35% increase), for the same spray status.

**Marks:** 1 for both ORs by exponentiating; 1 for both interpretations (multiplicative on the *odds*, other variable held fixed).

**Common errors:** interpreting as a change in probability; using $e^{-0.64442}$ as $1-0.64442$; saying "0.52 times more".

**(b)** *$\hat{p}$ and 95% CI for a sprayed plot with rain $= 8$.* [3]

With $c = (1,\, 8,\, 1)^T$:

$$\begin{aligned}
\hat{\eta} &= -1.67973 + 8(0.29876) - 0.64442 = 0.06593 \\
\mathrm{Var}(\hat{\eta}) &= c^T V c = V_{00} + 64V_{11} + V_{22} + 16V_{01} + 2V_{02} + 16V_{12} \\
&= 0.13212 + 0.18233 + 0.04824 - 0.28754 + 0.03635 - 0.09365 \\
&= 0.017844 \\
\mathrm{SE}(\hat{\eta}) &= \sqrt{0.017844} = 0.13358 \\
\hat{\eta} \pm 1.96\,\mathrm{SE} &= 0.06593 \pm 0.26182 = (-0.19589,\ 0.32775) \\
\hat{p} &= \frac{1}{1 + e^{-0.06593}} = \boxed{0.52} \\
p_{\mathrm{lo}} &= \frac{1}{1 + e^{0.19589}} = 0.4512 \\
p_{\mathrm{hi}} &= \frac{1}{1 + e^{-0.32775}} = 0.5812 \\
&\boxed{95\%\ \text{CI for } p:\ (0.45,\ 0.58)}
\end{aligned}$$

**FX-82:** Terms: $64 \times 0.002848834$, $16 \times (-0.017971204)$, $2 \times 0.018175543$, $16 \times (-0.005853401)$. Store the sum, press $\sqrt{}$. Inverse logit: $1 \div (1 + e^x(-)\,\hat{\eta})$.

**Marks:** 1 for $\hat{\eta}$ and $\hat{p}$; 1 for the variance including all covariance terms with correct multipliers; 1 for the CI on the $\eta$ scale transformed to $(0.45, 0.58)$.

**Common errors:** leaving out the covariance terms (then SE $= 0.60$ and the CI is far too wide); transforming SE rather than the endpoints; omitting the factor 2 or the 8 in cross terms; using $x = (1, 8, 0)$.

**(c)** *LRT for* ***spray*** *at the 5% level.* [2]

$$\begin{aligned}
&H_0 : \beta_2 = 0 \ \ \text{(rain only)} \qquad H_1 : \beta_2 \neq 0 \\
&D = D_0 - D_1 = 23.372 - 14.437 = 8.935 \approx \boxed{8.94} && \text{(difference in residual deviances)} \\
&D \overset{H_0}{\sim} \chi^2_1 \ \ (14 - 13 = 1\ \text{df}) && \text{(approx.)} \\
&8.94 > 3.84 && \text{(reject } H_0\text{)}
\end{aligned}$$

Conclusion: reject $H_0$ at the 5% level (printed $p = 0.0028$). Spray is needed in the model, even after rain is included.

**Marks:** 1 for hypotheses and $D = 8.94$ with $\chi^2_1$; 1 for comparing with 3.84 and the conclusion in context.

**Common errors:** using the wrong deviance order (negative $D$); *df* $= 13$ or 14; "accept $H_0$".

**(d)** *Is model 1 adequate by residual deviance? How might binomial assumptions fail?* [2]

$$\begin{aligned}
D_{\mathrm{res}} = 14.44 \text{ on } 13 \text{ df}, \qquad 14.44 < 22.36 = \chi^2_{13,0.95} \\
\frac{D_{\mathrm{res}}}{\mathrm{df}} = \frac{14.44}{13} = 1.11 \approx 1
\end{aligned}$$

Model 1 is adequate: no evidence of lack of fit ($D$ below the critical value). Binomial assumptions could fail because trees in the same plot share soil, microclimate and disease spread, so infections are positively correlated and not independent with a common $p$. This gives extra-binomial variation (overdispersion), so the binomial standard errors are too small, making CIs too narrow and $p$-values too small.

**Marks:** 1 for the comparison with 22.36 and "adequate"; 1 for a valid reason and the effect on SEs (too small).

**Common errors:** saying large deviance means good fit; saying SEs would be too large.

## Question 4 [10 marks]

**(a)** *Two pieces of evidence of overdispersion in model A.* [2]

$$\frac{D_{\text{res}}}{\text{df}} = \frac{222.13}{31} = 7.17 \gg 1$$
$$\hat{\phi}_A = \frac{X^2}{n-p} = 7.16976 \approx \boxed{7.17} \gg 1 \qquad \text{(Pearson estimate, should be} \approx 1 \text{ for Poisson)}$$

Also: the residual deviance 222.13 far exceeds the $\chi^2_{31}$ 95% point (44.99). (Note $\hat{\phi}_B = 6.75$ also $\gg 1$.)

**Marks:** 1 each for two valid quantitative pieces ($\hat{\phi}_A = 7.17$; deviance/df $\approx 7.2$ on 31 df).

**Common errors:** quoting only the null deviance; saying "$\hat{\phi} > 1$" without giving the value.

**(b)** *F-statistic for* `anova(modelD, modelC, test="F")`*, its df, and conclusion.* [3]

$$F = \frac{(D_B - D_A)/(\text{df}_B - \text{df}_A)}{\hat{\phi}_A} \qquad (\hat{\phi} \text{ from the larger model})$$
$$= \frac{(234.27 - 222.13)/(34-31)}{7.16976} = \frac{12.131/3}{7.16976} = \frac{4.0437}{7.16976}$$
$$= \boxed{0.56} \qquad \text{df} = \boxed{(3,\ 31)}$$

**FX-82:** $12.131 \div 3 \div 7.16976 = 0.5640$ (using $234.27 - 222.13 = 12.14$ gives 0.5651, same 2 d.p. answer, 0.56).

Since $0.56 < 2.91$, we do not reject $H_0$: `sst` and `zone` can be dropped (once overdispersion is allowed for).

**Marks:** 1 for $F$ formula with $\hat{\phi}_A$ in the denominator; 1 for $F = 0.56$; 1 for df $(3, 31)$ and conclusion.

**Common errors:** dividing by $\hat{\phi}_B$ or by 1; df $(3, 34)$; forgetting to divide the deviance change by 3.

**(c)** *Quasi-Poisson SE of* `depth`*, and 95% CI for the factor change in density per 10 m.* [3]

$$\text{SE}_{\text{quasi}} = \text{SE}_{\text{Poisson}}\sqrt{\hat{\phi}_A} \qquad (\text{SE scaled by } \sqrt{\hat{\phi}})$$
$$= 0.001689 \times \sqrt{7.16976} = 0.001689 \times 2.67764 = 0.004523 \approx \boxed{0.0045}$$
$$\hat{\beta} \pm t\,\text{SE} = -0.006366 \pm 2.04 \times 0.004523 = -0.006366 \pm 0.009227$$
$$= (-0.015593,\ 0.002861)$$
$$\text{10 m:}\ (-0.15593,\ 0.02861) \qquad (\text{multiply by 10})$$
$$\exp(\cdot):\ \left(e^{-0.15593},\ e^{0.02861}\right)$$
$$\boxed{(0.86,\ 1.03)} \qquad (\text{point estimate } e^{-0.06366} = 0.94)$$

**FX-82:** $\sqrt{7.16976}$ then $\times 0.001689$; then $e^x$ of $10 \times (-0.006366 \mp 0.009227)$.

The interval contains 1: no significant depth effect on density once overdispersion is allowed for.

**Marks:** 1 for SE = 0.0045; 1 for the interval for $\beta$ with $t = 2.04$; 1 for $\times 10$, exponentiating, and the result $(0.86, 1.03)$.

**Common errors:** using $\hat{\phi}$ instead of $\sqrt{\hat{\phi}}$; using the Poisson SE (gives the too-narrow $(0.91, 0.97)$); exponentiating without multiplying by 10; forgetting the interval is for the *factor*, not $\beta$.

**(d)** *How quasi-Poisson differs; why Chisq and F disagree; which to trust.* [2]

Model answer: Quasi-Poisson keeps the Poisson mean model (same $\hat{\beta}$) but assumes $\text{Var}(Y) = \phi\mu$ with $\phi$ estimated ($\hat{\phi} = 7.17$), so all SEs are multiplied by $\sqrt{\hat{\phi}}$ and tests use $F$ rather than $\chi^2$. The Chisq analysis of deviance assumes $\phi = 1$, so with $\hat{\phi} \approx 7$ it overstates the evidence ($p = 0.007$), while the $F$-test divides the deviance change by $\hat{\phi}$ and gives $p = 0.64$. Trust the $F$-test (the data are strongly overdispersed).

**Marks:** 1 for the Var $= \phi\mu$ difference (same $\hat{\beta}$, SEs scaled); 1 for explaining the disagreement via ignored overdispersion and choosing $F$.

**Common errors:** saying quasi-Poisson changes the coefficient estimates; trusting the Chisq test because its $p$-value is smaller.

## Question 5 [7 marks]

**(a)** $g'(\mu)$ *and show* $W_{ii} = \mu_i^2$. [1]

$$g(\mu) = \mu^{-1} \;\Rightarrow\; g'(\mu) = -\mu^{-2}$$
$$W_{ii} = \frac{1}{g'(\mu_i)^2\, v(\mu_i)} = \frac{1}{\mu_i^{-4}\,\mu_i^2} = \frac{1}{\mu_i^{-2}} = \mu_i^2$$

**Marks:** 1 for $g'(\mu) = -1/\mu^2$ and the simplification to $\mu_i^2$.

**Common errors:** $g'(\mu) = -1/\mu$; forgetting to square $g'$ in $W$; $v(\mu) = \mu$ (instead of $\mu^2$).

**(b)** *One IWLS step from* $\beta^0 = (0.45, 0.10)^T$*:* $\eta_i, \mu_i, z_i, W_{ii}$. [2]

Use $\eta_i = 0.45 + 0.10x_i$, $\mu_i = 1/\eta_i$, $z_i = \eta_i + (y_i - \mu_i)g'(\mu_i) = \eta_i - (y_i - \mu_i)/\mu_i^2$, $W_{ii} = \mu_i^2$.

$$i = 1: \;\; \eta_1 = 0.45,\; \mu_1 = 2.2222,\; y_1 - \mu_1 = 0.1778,\; \mu_1^2 = 4.9383$$
$$z_1 = 0.45 - \frac{0.1778}{4.9383} = 0.45 - 0.0360 = 0.4140$$
$$i = 2: \;\; \eta_2 = 0.55,\; \mu_2 = 1.8182,\; y_2 - \mu_2 = -0.1182,\; \mu_2^2 = 3.3058$$
$$z_2 = 0.55 + \frac{0.1182}{3.3058} = 0.55 + 0.0358 = 0.5858$$
$$i = 3: \;\; \eta_3 = 0.65,\; \mu_3 = 1.5385,\; y_3 - \mu_3 = -0.4385,\; \mu_3^2 = 2.3669$$
$$z_3 = 0.65 + \frac{0.4385}{2.3669} = 0.65 + 0.1853 = 0.8353$$

| $i$ | 1 | 2 | 3 |
|---|---|---|---|
| $\eta_i$ | 0.4500 | 0.5500 | 0.6500 |
| $\mu_i$ | 2.2222 | 1.8182 | 1.5385 |
| $z_i$ | 0.4140 | 0.5858 | 0.8353 |
| $W_{ii}$ | 4.9383 | 3.3058 | 2.3669 |

**FX-82:** $1 \div 0.45 =$ (use $x^{-1}$ key), then $\mu^2$ with $x^2$.

**Marks:** 1 for $\eta_i, \mu_i, W_{ii}$ correct; 1 for $z_i$ correct (sign of the correction term).

**Common errors:** using $z = \eta + (y - \mu)/\mu^2$ (wrong sign of $g'$); using $z = \eta + (y - \mu)$.

**(c)** *WLS formula for $\beta^1$; $X^TWX$, $X^TWz$, and $\beta^1$.* [2]

$$\beta^1 = (X^TWX)^{-1}X^TWz, \qquad X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$$

$$X^TWX = \begin{pmatrix} \sum W_i & \sum W_i x_i \\ \sum W_i x_i & \sum W_i x_i^2 \end{pmatrix}$$

$$\sum W_i = 4.9383 + 3.3058 + 2.3669 = 10.6109$$

$$\sum W_i x_i = 3.3058 + 2(2.3669) = 8.0395$$

$$\sum W_i x_i^2 = 3.3058 + 4(2.3669) = 12.7733$$

$$X^TWz = \begin{pmatrix} \sum W_i z_i \\ \sum W_i x_i z_i \end{pmatrix} = \begin{pmatrix} 2.0444 + 1.9364 + 1.9769 \\ 1.9364 + 2(1.9769) \end{pmatrix} = \begin{pmatrix} 5.9577 \\ 5.8902 \end{pmatrix}$$

$$\det = 10.6109(12.7733) - 8.0395^2 = 135.5358 - 64.6337 = 70.9021$$

$$\beta^1 = \frac{1}{70.9021}\begin{pmatrix} 12.7733 & -8.0395 \\ -8.0395 & 10.6109 \end{pmatrix}\begin{pmatrix} 5.9577 \\ 5.8902 \end{pmatrix} = \frac{1}{70.9021}\begin{pmatrix} 76.0996 - 47.3543 \\ -47.8972 + 62.5004 \end{pmatrix} = \frac{1}{70.9021}\begin{pmatrix} 28.7453 \\ 14.6032 \end{pmatrix}$$

$$\boxed{\beta^1 = (0.41,\ 0.21)^T} \qquad \text{(unrounded } (0.4054,\ 0.2060)\text{)}$$

**Marks:** 1 for the formula and correct $X^TWX$, $X^TWz$; 1 for solving the $2 \times 2$ system to $(0.41, 0.21)$.

**Common errors:** forgetting $W$ in $X^TWz$; inverse formula mistakes (sign of the off-diagonal); using $y$ instead of $z$.

**(d)** *Cook's distance for machine 8; what are machines 7 and 8?* [2]

$$C_8 = \frac{r_P^2\, h}{p\,\hat{\phi}\,(1-h)^2} \qquad (r_P = 0.223,\ h = 0.578,\ p = 2,\ \hat{\phi} = 0.1985)$$

$$= \frac{0.223^2 \times 0.578}{2 \times 0.1985256 \times 0.422^2} = \frac{0.049729 \times 0.578}{2 \times 0.1985256 \times 0.178084}$$

$$= \frac{0.028743}{0.070710} = \boxed{0.41}$$

**FX-82:** numerator $0.223^2 \times 0.578$; denominator $2 \times 0.1985256 \times 0.422^2$; divide.

Machine 7 (smallest Pearson residual $-0.794$) is an *outlier*: its cost (0.25) is much lower than the model predicts, but its leverage is modest (0.188), so $C_7 = 0.45$. Machine 8 (leverage $0.578 > 2p/n = 0.5$, age 14 far from the rest) is a *high-leverage* point, but it lies near the fitted curve (residual 0.223), so $C_8 = 0.41$. Neither $C_i$ approaches 1, so neither changes the fit much.

**Marks:** 1 for $C_8 = 0.41$ with correct formula; 1 for identifying 7 as outlier and 8 as high leverage and judging influence.

**Common errors:** using $h/(1-h)$ with $r_P^2$ (one factor of $1-h$ missing); omitting $\hat{\phi}$; calling a high-leverage point automatically influential.

## Question 6 [10 marks]

**(a)** *Derive* $Q(\theta, \theta^0)$. [4]

Let $\gamma_{jk} = P(Z_j = k \mid \mathbf{X}_j, \theta^0)$, $\pi_1 = \pi$, $\pi_2 = 1 - \pi$.

$$P(\mathbf{X}, \mathbf{Z} \mid \theta) = \prod_{j=1}^{6} \prod_{k=1}^{2} \Big[\pi_k \prod_{l=1}^{2} \phi(x_{jl}; \mu_k, 1)\Big]^{\mathbf{1}(Z_j = k)} \qquad \text{(units independent)}$$

$$\log P = \sum_j \sum_k \mathbf{1}(Z_j = k) \Big[\log \pi_k - \log 2\pi - \tfrac{1}{2} \sum_{l=1}^{2} (x_{jl} - \mu_k)^2\Big]$$

$$\mathbb{E}\left[\mathbf{1}(Z_j = k) \mid \mathbf{X}, \theta^0\right] = \gamma_{jk} \qquad \text{(E-step)}$$

$$\boxed{Q(\theta, \theta^0) = \sum_{j=1}^{6} \sum_{k=1}^{2} \gamma_{jk} \Big[\log \pi_k - \tfrac{1}{2} \sum_{l=1}^{2} (x_{jl} - \mu_k)^2 - \log 2\pi\Big]}$$

**Marks:** 1 for the complete-data likelihood (both measurements share $Z_j$); 1 for its log; 1 for replacing indicators by $\gamma_{jk}$; 1 for the final $Q$.

**Common errors:** treating the two measurements as having separate $Z$s; only one $(x - \mu)^2$ term per unit; $\gamma$ not depending on $j, k$.

**(b)** *M-step updates for* $\pi$ *and* $\mu_1$. [3]

$$\frac{\partial Q}{\partial \pi} = \frac{\sum_j \gamma_{j1}}{\pi} - \frac{\sum_j \gamma_{j2}}{1 - \pi} = 0$$

$$(1 - \pi) \sum_j \gamma_{j1} = \pi \sum_j \gamma_{j2}$$

$$\sum_j \gamma_{j1} = \pi \sum_j (\gamma_{j1} + \gamma_{j2}) = 6\pi \qquad (\gamma_{j1} + \gamma_{j2} = 1)$$

$$\boxed{\hat{\pi} = \frac{1}{6} \sum_{j=1}^{6} \gamma_{j1}}$$

$$\frac{\partial Q}{\partial \mu_1} = \sum_j \gamma_{j1} \sum_{l=1}^{2} (x_{jl} - \mu_1) = 0$$

$$\sum_j \gamma_{j1} (x_{j1} + x_{j2}) = 2\mu_1 \sum_j \gamma_{j1}$$

$$\boxed{\hat{\mu}_1 = \frac{\sum_j \gamma_{j1} (x_{j1} + x_{j2})}{2 \sum_j \gamma_{j1}}}$$

(Second derivatives $-\sum \gamma/\pi^2 - \cdots < 0$ and $-2\sum_j \gamma_{j1} < 0$, so these are maxima.)

**Marks:** 1 for $\hat{\pi}$; 1 for the derivative setup for $\mu_1$; 1 for $\hat{\mu}_1$ with the factor 2 (two measurements per unit).

**Common errors:** dividing by $\sum_j \gamma_{j1}$ instead of $2\sum_j \gamma_{j1}$ while summing both $x$'s; dividing $\hat{\pi}$ by 12.

**(c)** $\gamma_{11}$ *for* $\theta^0 = (0.4, 0, 3)$, $\mathbf{x}_1 = (1.5, 2.5)$. [2]

$$\gamma_{11} = \frac{\pi^0 f(\mathbf{x}_1 \mid \mu_1^0)}{\pi^0 f(\mathbf{x}_1 \mid \mu_1^0) + (1-\pi^0) f(\mathbf{x}_1 \mid \mu_2^0)} \qquad \text{(Bayes' rule)}$$

$$f(\mathbf{x}_1 \mid \mu_1^0) \propto \exp\Big\{ -\tfrac{1}{2}(1.5^2 + 2.5^2) \Big\} = e^{-4.25} \qquad (\mu = 0)$$

$$f(\mathbf{x}_1 \mid \mu_2^0) \propto \exp\Big\{ -\tfrac{1}{2}\big[(1.5-3)^2 + (2.5-3)^2\big] \Big\} = e^{-1.25} \qquad (\mu = 3;\ \text{common } (2\pi)^{-1} \text{ cancels})$$

$$\gamma_{11} = \frac{0.4e^{-4.25}}{0.4e^{-4.25} + 0.6e^{-1.25}} = \frac{1}{1 + 1.5e^{3}}$$

$$= \frac{1}{1 + 1.5 \times 20.0855} = \frac{1}{31.1283} = 0.0321 \ \Rightarrow\ \boxed{0.03}$$

**FX-82:** $e^x\,3 = 20.0855$; $\times 1.5 = 30.1283$; $+1$; then $x^{-1}$. Keep 0.0321 for part (d).

**Marks:** 1 for the Bayes' rule set-up with correct exponents $-4.25$ and $-1.25$; 1 for the value $0.0321 \approx 0.03$.

**Common errors:** using only one measurement; using $\mu_2 = 0$ and $\mu_1 = 3$ swapped; forgetting the priors $0.4, 0.6$; $\sigma^2$ in the exponent mis-handled.

**(d)** *Updated* $\hat{\pi}$ *and* $\hat{\mu}_1$ *using the given* $\gamma_{j1}$. [1]

Sums $s_j = x_{j1} + x_{j2} = (4,\ 1.5,\ 4.4,\ 3.4,\ 0.3,\ 4.1)$ and
$\gamma_{j1} = (0.0321,\ 0.9836,\ 0.0099,\ 0.1672,\ 0.9995,\ 0.0240)$.

$$\sum_j \gamma_{j1} = 0.0321 + 0.9836 + 0.0099 + 0.1672 + 0.9995 + 0.0240 = 2.2163$$

$$\hat{\pi} = \frac{2.2163}{6} = 0.3694 = \boxed{0.37}$$

$$\sum_j \gamma_{j1} s_j = 0.1284 + 1.4754 + 0.0436 + 0.5685 + 0.2999 + 0.0984 = 2.6142$$

$$\hat{\mu}_1 = \frac{2.6142}{2 \times 2.2163} = \frac{2.6142}{4.4326} = 0.5897 = \boxed{0.59}$$

**Marks:** 1 for both $\hat{\pi} = 0.37$ and $\hat{\mu}_1 = 0.59$ (follow-through from your $\gamma_{11}$).

**Common errors:** using $\sum \gamma s / \sum \gamma$ (no factor 2, giving 1.18); not pairing $\gamma_{j1}$ with the right $s_j$.

## Question 7 [17 marks]

Summary statistics used throughout: $\sum x_i^2 = 1 + 4 + 9 = 14$, $\sum x_i y_i = 1.5 + 4 + 10.5 = 16$, $\sum y_i^2 = 2.25 + 4 + 12.25 = 18.5$.
Joint posterior (up to a constant), with $S(\beta) = \sum_i (y_i - \beta x_i)^2$:

$$p(\beta, \tau \mid \mathbf{y}) \propto \underbrace{\tau^{3/2} e^{-\frac{\tau}{2} S(\beta)}}_{\text{likelihood}} \underbrace{\tau^{1/2} e^{-\frac{\tau}{2}\beta^2}}_{p(\beta \mid \tau)} \underbrace{\tau^2 e^{-2\tau}}_{p(\tau)} = \tau^4 \exp\Big\{ -\tau \Big[ 2 + \tfrac{1}{2}\big(S(\beta) + \beta^2\big) \Big] \Big\}$$

**(a)(i)** $p(\beta \mid \tau, \mathbf{y})$. [3]

$$p(\beta \mid \tau, \mathbf{y}) \propto \exp\Big\{ -\tfrac{\tau}{2}\big[S(\beta) + \beta^2\big] \Big\} \qquad \text{(drop terms free of } \beta\text{)}$$

$$= \exp\Big\{ -\tfrac{\tau}{2}\Big[ \beta^2\big(\textstyle\sum x_i^2 + 1\big) - 2\beta \sum x_i y_i + \sum y_i^2 \Big] \Big\}$$

$$\propto \exp\Big\{ -\tfrac{\tau}{2}\big[15\beta^2 - 32\beta\big] \Big\}$$

$$\propto \exp\Big\{ -\tfrac{15\tau}{2}\Big(\beta - \tfrac{16}{15}\Big)^2 \Big\} \qquad \text{(complete the square)}$$

$$\boxed{\beta \mid \tau, \mathbf{y} \sim \text{Normal}\Big(\tfrac{16}{15} = 1.07,\ \text{variance } \tfrac{1}{15\tau}\Big)}$$

**Marks:** 1 for combining likelihood and prior in the exponent; 1 for completing the square; 1 for Normal with mean 16/15 and precision $15\tau$.

**Common errors:** forgetting the prior's +1 (using 14 and 16/14); variance 1/15 without $\tau$; treating the $\tau$ factors as part of the kernel for $\beta$.

**(a)(ii)** $p(\tau \mid \beta, \mathbf{y})$. [3]

$$p(\tau \mid \beta, \mathbf{y}) \propto \tau^{3/2+1/2+2} \exp\Big\{ -\tau\Big[2 + \tfrac{1}{2}\big(S(\beta) + \beta^2\big)\Big]\Big\} \qquad \text{(keep factors in } \tau\text{)}$$

$$= \tau^{5-1} \exp\{-\tau \cdot \text{rate}\} \qquad \text{(Gamma kernel } \tau^{a-1}e^{-b\tau}\text{)}$$

$$\boxed{\tau \mid \beta, \mathbf{y} \sim \text{Gamma}\Big(5,\ 2 + \tfrac{1}{2}\Big[\textstyle\sum_i (y_i - \beta x_i)^2 + \beta^2\Big]\Big)}$$

**Marks:** 1 for powers of $\tau$ collected: 3/2 (likelihood, $n/2$), 1/2 (prior on $\beta$), 2 (prior on $\tau$); 1 for shape 5; 1 for the rate.

**Common errors:** shape $3 + 3/2 = 4.5$ (forgetting the 1/2 from $p(\beta \mid \tau)$); forgetting the $\beta^2$ in the rate; using the rate 2 only.

**(b)** *Acceptance probability at* $(\beta_c, \tau_c) = (1, 2)$, $(\beta_n, \tau_n) = (1.2, 3)$. [6]

$$\alpha = \min\left\{1,\ \frac{\pi(\beta_n, \tau_n)}{\pi(\beta_c, \tau_c)} \cdot \frac{q(\tau_c \mid \tau_n)}{q(\tau_n \mid \tau_c)}\right\} \qquad (\pi\text{=posterior kernel; } \beta \text{ proposal symmetric})$$

**Posterior ratio.** $S(1) = 0.5^2 + 0^2 + 0.5^2 = 0.5$; $S(1.2) = 0.3^2 + (-0.4)^2 + (-0.1)^2 = 0.26$.

$$\text{current: } 2 + \tfrac{1}{2}(0.5 + 1^2) = 2.75$$

$$\text{new: } 2 + \tfrac{1}{2}(0.26 + 1.2^2) = 2.85$$

$$\frac{\pi_n}{\pi_c} = \frac{3^4 e^{-3(2.85)}}{2^4 e^{-2(2.75)}} = \left(\frac{3}{2}\right)^4 e^{-8.55+5.5}$$

$$= 5.0625 \times e^{-3.05} = 5.0625 \times 0.04736 = 0.2398$$

**Proposal ratio.** The $\tau$-proposal is Gamma(2, rate $2/\tau_c$): density $r^2\tau e^{-r\tau}$.

$$q(\tau_n \mid \tau_c):\ r = 2/2 = 1,\ \tau = 3:\ 1^2 \cdot 3e^{-3} = 0.14936$$

$$q(\tau_c \mid \tau_n):\ r = 2/3,\ \tau = 2:\ \left(\tfrac{2}{3}\right)^2 \cdot 2\, e^{-4/3} = 0.23431$$

$$\frac{q(\tau_c \mid \tau_n)}{q(\tau_n \mid \tau_c)} = \frac{0.23431}{0.14936} = \frac{8}{27} e^{5/3} = 1.5687$$

$$\alpha = \min\{1,\ 0.2398 \times 1.5687\} = \min\{1,\ 0.3761\} = \boxed{0.38}$$

**FX-82:** $e^x(-)\,3.05 = 0.04736$; $\times 5.0625$. Proposal: $8 \div 27 \times e^x(5 \div 3)$. Multiply the two ratios.

**Marks:** 1 for the correct MH ratio incl. the proposal correction; 1 for noting $\beta$ cancels (symmetric Normal); 2 for the posterior ratio (kernel and arithmetic); 1 for the proposal ratio; 1 for $\alpha = 0.38$.

**Common errors:** using the Gibbs-conditional instead of the joint kernel; omitting $q$ ratio (gives 0.24); inverting the $q$ ratio (gives 0.15); using the scale rather than the rate $2/\tau_c$ for the Gamma proposals; forgetting prior and $\tau^{1/2}$ from $p(\beta \mid \tau)$.

**(c)(i)** *Optimal $q^*(\tau)$ in terms of $\mathbb{E}_q[\beta], \mathrm{Var}_q[\beta]$.* [4]

$$\begin{aligned}
\log q^*(\tau) &= \mathbb{E}_{q(\beta)}\big[\log p(\mathbf{y},\beta,\tau)\big] + \text{const} && \text{(CAVI)}\\
&= 4\log\tau - \tau\Big[2 + \tfrac{1}{2}\big(\mathbb{E}_q[S(\beta)] + \mathbb{E}_q[\beta^2]\big)\Big] + \text{const}\\
\mathbb{E}_q[S(\beta)] &= \sum y_i^2 - 2\sum x_i y_i\ \mathbb{E}_q[\beta] + \sum x_i^2\ \mathbb{E}_q[\beta^2]\\
\mathbb{E}_q[\beta^2] &= \mathrm{Var}_q[\beta] + \mathbb{E}_q[\beta]^2 && \text{(second moment)}\\
\mathbb{E}_q[S] + \mathbb{E}_q[\beta^2] &= 18.5 - 32\ \mathbb{E}_q[\beta] + 15\big(\mathrm{Var}_q[\beta] + \mathbb{E}_q[\beta]^2\big)\\
&\boxed{q^*(\tau) = \mathrm{Gamma}(5,\ b^*)},\ \text{where}\\
b^* &= 2 + \tfrac{1}{2}\Big[18.5 - 32\ \mathbb{E}_q[\beta] + 15\big(\mathrm{Var}_q[\beta] + \mathbb{E}_q[\beta]^2\big)\Big]
\end{aligned}$$

**Marks:** 1 for $\mathbb{E}_q$ of the log joint; 1 for expanding $\mathbb{E}_q[S]$; 1 for $\mathbb{E}_q[\beta^2] = \mathrm{Var} + \mathbb{E}^2$; 1 for Gamma with shape 5 and correct rate.

**Common errors:** replacing $\mathbb{E}_q[\beta^2]$ by $\mathbb{E}_q[\beta]^2$ (missing $\mathrm{Var}_q$); forgetting the prior's $\beta^2$ term (using 14 not 15); shape 4.

**(c)(ii)** $\mathbb{E}_q[\beta] = 1.1$, $\mathrm{Var}_q[\beta] = 0.02$*: parameters of $q^*$ and $\mathbb{E}_{q^*}[\tau]$.* [1]

$$\begin{aligned}
\mathbb{E}_q[\beta^2] &= 0.02 + 1.1^2 = 1.23\\
\text{rate} &= 2 + \tfrac{1}{2}\big[18.5 - 32(1.1) + 15(1.23)\big] = 2 + \tfrac{1}{2}\big[18.5 - 35.2 + 18.45\big] = 2 + \tfrac{1}{2}(1.75) = 2.875\\
\mathbb{E}_{q^*}[\tau] &= \frac{5}{2.875} = \boxed{1.74} \qquad \Big(q^* = \mathrm{Gamma}(5,\ 2.88)\Big)
\end{aligned}$$

**Marks:** 1 for rate 2.875 (shape 5) and $\mathbb{E}[\tau] = 1.74$.

**Common errors:** arithmetic slip with $32 \times 1.1$; using the scale instead of the rate for the mean ($5 \times 2.875$).

## Question 8 [5 marks]

**(a)** *Smallest $M$ with $f(x) \le Mg(x)$.* [3]

$$\begin{aligned}
\frac{f(x)}{g(x)} &= \frac{xe^{-x^2/2}}{e^{-x}} = x\,e^{x - x^2/2} =: r(x)\\
\frac{d}{dx}\log r &= \frac{1}{x} + 1 - x = 0 && \text{(maximise } \log r(x) = \log x + x - x^2/2\text{)}\\
x^2 - x - 1 = 0 \ \Rightarrow\ x^* &= \frac{1+\sqrt{5}}{2} = 1.6180 && \text{(positive root; } (\log r)'' < 0\text{)}\\
M = r(x^*) &= 1.6180\, e^{1.6180 - 1.3090} && (x^{*2} = x^* + 1 = 2.618\text{, half is } 1.309)\\
&= 1.6180 \times e^{0.3090} = 1.6180 \times 1.3621\\
&\boxed{M = 2.20}
\end{aligned}$$

**FX-82:** $(1+\sqrt{5}) \div 2 = 1.61803$; $e^x(1.61803 - 1.30902) = 1.36219$; multiply.

**Marks:** 1 for $f/g = xe^{x - x^2/2}$; 1 for solving for $x^* = 1.618$; 1 for $M = 2.20$ (with justification of max, as $r \to 0$ at 0 and $\infty$).

**Common errors:** maximising $f$ alone (gives $x = 1$, $M = 1.65$); wrong root; using $M = \max f = 0.607$.

**(b)** *What does the last line do? Output for $x = 0.5$, $u = 0.5$.* [1]

It accepts the proposal $x$ (returns it) when $u < \dfrac{f(x)}{M\,g(x)}$, and returns `NA` (reject) otherwise.

$$\frac{f(0.5)}{Mg(0.5)} = \frac{0.5\,e^{0.5-0.125}}{2.2039} = \frac{0.7275}{2.2039} = 0.3301$$
$$u = 0.5 > 0.3301 \qquad \text{(reject)}$$

The code returns $\boxed{\texttt{NA}}$.

**Marks:** 1 for correctly describing the accept test and getting `NA` with the numbers.

**Common errors:** returning 0.5; comparing $u$ with $f(x)/g(x)$ without dividing by $M$.

**(c)** *Probability one attempt returns a sample; mean attempts per sample.* [1]

$$P(\text{accept}) = \int_0^\infty \frac{f(x)}{Mg(x)}\,g(x)\,dx = \frac{1}{M}\int_0^\infty f(x)\,dx = \frac{1}{M} \qquad (f \text{ is a normalised density})$$
$$= \frac{1}{2.2039} = \boxed{0.45}$$
$$\text{mean attempts} = \frac{1}{P(\text{accept})} = M = \boxed{2.20} \qquad \text{(geometric)}$$

**Marks:** 1 for 0.45 and 2.20 (both).

**Common errors:** $P(\text{accept}) = M$; quoting mean attempts as 0.45.

## Question 9 [5 marks]

**(a)** *Burn-in from the trace plot; why discard.* [1]

Discard about the first $\boxed{100}$ iterations (chain B has merged with chain A by about iteration 50–100; any answer in 50–100 is fine). Reason: early draws depend on the arbitrary starting values ($-8$, 18) and are not from the stationary (target) distribution, so keeping them biases the estimates.

**Marks:** 1 for a value in the range 50–100 and a reason about starting values/not yet stationary.

**Common errors:** discarding 300 or more without justification; saying burn-in "removes autocorrelation".

**(b)** $\hat{R} = B/W$ *for iterations 1–300 and 301–1200.* [2]

$$\text{1–300: } W = \frac{2.65 + 3.23}{2} = 2.94, \quad B = 3.36$$
$$\hat{R} = \frac{B}{W} = \frac{3.36}{2.94} = 1.1429 = \boxed{1.14} > 1.05$$
$$\text{301–1200: } W = \frac{2.29 + 2.28}{2} = 2.285, \quad B = 2.29$$
$$\hat{R} = \frac{2.29}{2.285} = 1.0022 = \boxed{1.00} < 1.05$$

Interpretation: for iterations 1–300 the pooled spread exceeds the within-chain spread because the chains started far apart (burn-in is included): not converged. For 301–1200 the chains agree: converged.

**Marks:** 1 for both $\hat{R}$ values; 1 for the interpretations against the 1.05 threshold.

**Common errors:** $W/B$ (gives 0.88); using the pooled sd as $W$; averaging the variances rather than the sds.

**(c)** *ESS and Monte Carlo SE; factor by which the naive SE is too small.* [2]

$$\text{ESS} = n\,\frac{1-\hat\rho_1}{1+\hat\rho_1} = 900 \times \frac{1-0.92}{1+0.92} = 900 \times \frac{0.08}{1.92} = \boxed{37.50}$$

$$\text{MCSE} = \frac{s}{\sqrt{\text{ESS}}} = \frac{2.29}{\sqrt{37.5}} = \frac{2.29}{6.1237} = \boxed{0.37}$$

$$\text{naive SE} = \frac{2.29}{\sqrt{900}} = \frac{2.29}{30} = 0.0763$$

$$\text{factor} = \frac{\text{MCSE}}{\text{naive}} = \sqrt{\frac{n}{\text{ESS}}} = \sqrt{\frac{1+\hat\rho_1}{1-\hat\rho_1}} = \sqrt{24} = \boxed{4.90}$$

**FX-82:** $900 \times 0.08 \div 1.92 = 37.5$; $2.29 \div \sqrt{37.5} = 0.374$; $\sqrt{24} = 4.899$.

**Marks:** 1 for ESS = 37.50; 1 for MCSE = 0.37 and the factor 4.90 (the naive SE is about 4.9 times too small).

**Common errors:** ESS $= n(1+\rho)/(1-\rho)$ (inverted); dividing $s$ by ESS instead of $\sqrt{\text{ESS}}$; using $\rho = 0.92$ as a fraction of the SE.